14. Stack Depth

Varying stack depth presents a whole new problem by distorting the thought processes we've built into our arsenal for playing 100BBs deep. It's not that everything changes, but there are some new implications to playing at non-standard stack depth that will cost us a lot of EV left unrecognised. We've all had to deal with the table where we're a terrifying 300BB deep with the solid aggressive player to our left (probably leave!) or when we have that 16BB Fish behind us irritatingly shoving over our every open. This is a short chapter in which we'll address how to handle these situations and everything in between.

Being forever uncomfortable both shallow and deep is not acceptable for the serious poker player and an approach of just avoiding these stack sizes is totally unsatisfactory. There is simply too much EV that can be pillaged in favourable non-standard stack situations. So let's get adapting.

14.1 Playing Deep Pre-Flop

Deep Stacks = More Implied Odds

This is the first important consideration for playing deep pre-flop. Hero is very often able to widen up his ranges for both flatting opens and flatting 3-bets. As always, when implied odds are high he should favour flatting hands that have a lot of **nut potential (NP)** such as pocket pairs, suited connectors and suited aces. We looked at this attribute when we considered starting hands way back in Figure 4. These NP hands, which usually connect too infrequently in a huge way at normal stack depth, can easily become calls where stacks are deeper and the reward is larger. After all, remember that implied odds are just a measure of Hero's investment to see the next street compared with what he stands to gain from Villains' stacks should he connect in the way he desires.

Take the following hand as an illustration:

Hand 138

UTG – Aggro Fish (36/18/10/25)
HJ – Unknown
CO – Hero
BU – Unknown
SB – Unknown
BB – Unknown

Dealt to Hero:

UTG raises to 3BB, HJ folds, **Hero?**

This is a spot we'd normally fold fairly quickly 100 BB deep. **86s** is not a hand that will flop something powerful very often. To recoup enough money on average to compensate for the 3BB investment it costs to see a flop here is not normally possible. If Hero knew he was unlikely to get squeezed and would indeed see the flop close to 100% of the time that he flatted, then calling would be more reasonable, but with three unknown players left to act behind and without the BU, calling is undesirable.

Let's look at what changes 200BB deep. With these much deeper stacks, Hero can potentially win a lot more from the Fish by flatting in position. He will have the option of making large flop raises, big turn bets and huge river bets until close to 200BBs are in the middle. This makes the ratio of investment to potential gain far better and Hero should definitely flat this spot even with knowledge that he'll get squeezed some relevant amount of the time.

So deeper stacks lessen the requirements for how often our hand needs to flop big in order to call. We can mine with weaker implied odds hands like **86s** when deep vs. a weaker player in position, but we'd need to be against a particularly horrible player and be very likely to see the flop after investing our 3BBs to call a hand as bad as **65o** in this spot.

Hero folds at 100BB stack depth.

Hero calls at 200BB stack depth.

Let's take an example where we face a 3-bet now.

Hand 139

UTG – Unknown
HJ – Unknown
CO – Hero
BU – Unknown
SB – Reg (23/17/**8**)
BB – Hero

Dealt to Hero:

UTG folds, HJ folds, **Hero raises to 2.5BB**, BU folds, **SB raises to 9BB**, BB folds, **Hero?**

Another hand that just shouldn't be a part of our defending range with 100BB stacks. It flops terribly the vast majority of the time and so we'll be forced to fold on a huge amount of flops or else adopt an insanely unbalanced floating/raising strategy. It's also an awful hand to 4-bet bluff since it blocks none of Villain's 3-bet value range whatsoever.

With 200BB stacks however, it's less fatal that we have to fold on a lot of flops. Those times we do flop a set (around 1/9) we'll be able to extract a lot more money when Villain is stacking off. This lessens the burden on how often we need to defend to a c-bet on the flop and improves implied odds

drastically.

Have a look back at the set mining rule we devised back in Chapter 6. We normally need to recoup roughly 10x our investment here in order to break even on a set mine assuming that we have little intention of playing anything but fit-or-fold post-flop, which is pretty much true with a hand as rigid as **44** - it either flops the world or nothing at all. We invest 6BB here to call the 3-bet and the pot will be 19BB if we do. We'll have to make 60BB in total to break even on a call, which equates to another 41BB from Villain's stack on top of the pot. Winning just under half of Villain's stack with 100BB stacks is optimistic, but when we can sometimes win 200BB, this task becomes much easier. This hand is fine to call with deeper stacks especially in position.

Hero folds at 100BB stack depth.

Hero calls at 200BB stack depth.

You may wonder at this point where the cut off lies in these two hands. At what stack size would our implied odds cease to be sufficient in each case? The answer is somewhere in the middle and it's really just a case of estimation. These spots are too complicated to calculate exactly due to the vast array of post-flop possibilities, but I'd say in the first case we should start calling at around 140BB stack depth and in the second about 160BB. What is this based off of? Experience, feel and an implicit familiarity with the post-flop possibilities, which are just too numerous to consciously or explicitly compute.

Practice this kind of estimation to develop a better feel for when stack depth causes us to call or fold hands we might not normally.

Stack Depth Magnifies Other Advantages

In Hand 139, at 200BB stack depth, we were able to make a call that would normally be -EV (worse than folding). This was due to the deeper stacks bolstering our implied odds and holding a hand with great nut potential. Implied odds is not the only weapon made more dangerous by deep stacks. Skill edge and positional edge are two other assets boosted by having more money behind. This can make bloating large pots out of position very undesirable without a significant equity edge to compensate, especially vs. competent opposition that can use this advantage effectively.

This means that 3-betting thinly for value deep is unadvisable in many situations where it'd otherwise be acceptable. Take the following hand, for example:

Hand 140

UTG – Unknown
HJ – Unknown
CO – Winning Reg (26/21/45/53) [250BB]
BU – Unknown
SB – Unknown
BB – Hero

Dealt to Hero:

UTG folds, HJ folds, **CO raises to 3BB**, BU folds, SB folds, **Hero?**

QQ is a hand that, 100BB deep, we'd normally be thrilled to have in our 3-bet and get it in range vs. any standard or looser Reg in these positions - it's a no brainer and must be part of our 3-bet/shove strategy.

At the stack size in question, however, there are two central reasons as to why 3-betting is not so desirable.

- As we saw previously, the opener's implied odds are better and he can very profitably defend much more of his opening range. This EV has to come from somewhere, and it's not being generously injected by the poker site, that's for sure! Hero is the one who grants Villain this extra EV by having to play a lot of good but non-nutted hands out of position post-flop in a large pot with a very large effective stack. What Villain gains in implied odds is equal to what Hero suffers in reverse implied odds.
- Villain's positional advantage and any skill advantage are magnified by the stack depth. This means that many decision points post-flop will be negative or at least lower EV for Hero that wouldn't ordinarily be at a smaller stack depth.

Calling is going to be the highest EV line for Hero in this situation given how unfavourable stacks are for 3-betting. Hero will be able to control the pot size much better and prevent Villain from being able to threaten anywhere near 200BBs. To do this Hero will simply adopt a defensive pyramidal calling strategy as we saw back in Chapter 8 on the boards where his range is suffering a lot against Villain's. The smaller pot then makes Villain's magnified positional advantage from stack depth less severe as Villain will struggle to grow pots to any size that makes the 200BB stack relevant without adopting some kind of bizarre over-betting strategy.

The potentially shocking conclusion for Hero is that he has no 3-bet range in this spot. The situation just favours the opener, not the 3-bettor.

Hero calls 3BB.

Some players might have two objections here:

- The first is about losing value by keeping the pot so small. Can we really play **AA** by just flatting?
- The second concerns Hero's lack of re-stealing; doesn't it make opening the pot with 100% of hands insanely profitable for villain?

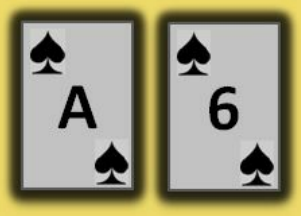
To the first objection I would respond by stressing that building huge pots 200BB deep does not actually favour overpairs; these hands cease to be sledgehammers that want to shovel in three large bets in a 3-bet pot vs. most opponents. Ranges for committing large amounts of money this deep are more polarised than this and so overpairs will frequently be bluff catchers. Hero will end up wanting to pot control a street post-flop anyway a large majority of the time even with a hand like **QQ** on **874r**. This is much more achievable by keeping the pot smaller. Of course if Villain is stationy then Hero should develop a 3-bet range and embrace the variation that ensues.

In response to the second objection, I would remind the reader that we are developing an optimal strategy vs. a balanced BU opening range that can continue to plenty of 3-bets profitably. If Villain makes the adjustment of opening very wide indeed, we can start to develop a polarised 3-bet range and punish him for having too many hands that can't call, even at this stack depth.

Being out of position deep is a nightmare vs. anyone competent, but being in position deep as the 3-bettor yields a much higher EV return and Hero should not pass up the opportunity to have a 3-bet range in these spots. It is then Hero who wields the magnified positional advantage and so he has no problem building big pots in which he can swing this weapon. Pot control, value, fold equity etc. are all in Hero's favour and so in the following hand, he can afford to play a much less timid pre-flop game.

Hand 141

UTG – Unknown
HJ – Unknown
CO – Winning Reg (20/16/57/52) [250BB]
BU – Hero
SB – Unknown
BB – Unknown

Dealt to Hero: 

UTG folds, HJ folds, **CO raises to 3BB**, **Hero?**

In this spot we can breathe a sigh of relief due to our glorious position and revert to a balanced polarised range against this player who folds a slightly higher than balanced percentage to 3-bets after raising. In fact, **A6s** is a particularly suitable choice to form 4 combos of our bluff portion. It's slightly too weak to flat (depending on who is in the blinds) it blocks Villain's continuing range, but more importantly at this stack depth, it can flop the nuts and cooler Villain for a very large amount of BBs.

Hero raises to 9.5BB.

Hero's sizing is larger than what we're used to (see Figure 62 for a reminder of what standard 3-bet sizing looks like). Hero benefits from using bigger sizing at this stack depth for two reasons:

- Hero gets to cut down Villain's implied odds and stop a large part of his range being very easy to defend.
- Hero does not need to fear this larger 3-bet size costing him too much EV when Villain 4-bets as Hero can flat a lot of his range vs. a 4-bet in this position. Villain could elect to 4-bet larger than usual, but this will cut back the profitability of his 4-betting by increasing RFE.

To conclude the first section of this chapter we've learnt two crucial things:

Deeper stacks enhance implied odds and make it much more profitable to be the pre-flop caller. This allows the caller to flat a wider range vs opens, 3-bets and 4-bets.

And:

Being out of position is not a *sufficient* reason to abandon having a 3-bet range.

Being very deep is not a *sufficient* reason to abandon having a 3-bet range.

Being out of position and being very deep can be a *sufficient* reason to abandon having a 3-bet range, especially where Hero is readless and Villain is competent.

It's now time to look at what happens when stacks are smaller than 100BB.

14.2 Playing Shallow Pre-Flop

There are a few different bands of effective stack size that fall under the umbrella term: 'shallow stacked'. We'll start off by examining the lowest stack size we're likely to encounter in a cash game of 10-20BB and then move upwards from there to 40BB stacks and finally 60-80BB stacks. Hero can then put the pieces together and draw his own conclusions about those stack sizes in between that weren't explicitly covered by the examples in this section.

10-20BB - The Pest Stack

A stack of 10-20BB waiting to act behind us can be a real nuisance and is the poker equivalent of that really annoying mosquito that dives in and out of your ears as you try to sleep. You know it won't do you any real long-term harm, but that doesn't stop it from ruining your night. Take the following example for instance and try to spot Hero's mistake.

Hand 142

UTG – Unknown
HJ – Unknown
CO – Hero
BU – Unknown
SB – Passive Fish (42/8/2/66)
BB – Unknown [17BB]

Dealt to Hero:

UTG folds, HJ folds, **Hero raises to 3BB**, BU folds, SB folds,
BB raises to 17BB (all in), **Hero folds**.

The mistake is not that Hero should refrain from opening a hand as good as **98s** from the CO, unless the BB is shoving a very wide range indeed, but Hero has no reason to believe this the case as of yet. So opening is still correct especially with the other fully stacked Fish out of position to Hero in the SB. It's the sizing that's the problem here. Hero's aims in opening are:

- To win the blinds.

- To play a pot in position with the initiative and a skill advantage 100BB deep vs. the SB.
- To play a three-way pot with a nice multiway hand in position vs. two weaker players.

How do we know that the BB is a weaker player? Because he has 17BB! Regs simply don't have 17BB unless they're professional short stackers and even these will usually auto reload to a slightly deeper stack depth. Moreover, short stacking Regs are a much rarer breed than random 17BB stacked Fish.

Anyway, the three aims above can all be accomplished with a **2BB** open. Sure, Hero loses a bit of value that way from the building a smaller pot vs. the fit-or-fold Fish in the SB, but he loses far more by 3xing and having to fold quite often to a shove. 10-20BB stacks will usually shove more often than a deeper stack will 3-bet. Just as Hero is advised to open smaller vs. habitual 3-bettors, he should do the same vs. habitual shovers. This lower part of his opening range that he has to fold will save a lot of money in the long run and the hands in this sub-range are his most common holdings.

It should also be pointed out that, in this spot, we don't need to worry about punishing BB if he does choose to flat wide with his tiny stack as many horrible players might. The effective stack naturally negates his implied odds rendering a wide flatting range a horrible strategy. Hero needn't open 3x even if he knows BB flats more than he should. Saving money against the shoves is still of paramount importance and so Hero should open his whole range for 2x in this situation.

Open smaller when shovable stacks wait behind. Money saved from losing less to a shove with hands we have to fold will almost always outweigh any benefit gained from using bigger sizing.

Where those stacks are known to be shoving very wide, stop opening hands with insufficient equity to call vs. Villain's shoving range.

Now we'll examine Hero's decision to fold to the shove having already made his sizing mistake. Remember that this spot is an end of action spot (if Hero calls there will be no more action for the hand). This means that he can simply evaluate whether his hand is likely to meet its RE vs. Villain's range.

RE = ATC / (ATC + TP)

RE = 14 / (14 + 20.5) = 40.5%

Using a reasonable assumption that Villain is shoving 20% of hands in this spot on average, Hero will have 37% equity and so folding is correct as played.

Now how should we approach the situation where the tiny stack opens and we're left with an oversized 3-bet remaining in our stack?

Hand 143

UTG – Unknown
HJ – Unknown
CO – Fish (29/16) [20BB] (20 hands)
BU – Unknown
SB – Hero
BB – Unknown

Dealt to Hero:

UTG folds, HJ folds, **CO raises to 3BB**, BU folds, **Hero?**

This is a spot where having a flatting range is just a categorical mistake. There is very little to be gained from playing in a fit-or-fold manner that tries to harvest implied odds as the payout is tiny compared to the investment, which represents more than 1/7th of the effective stack. There is, however, a large amount of EV to be gained from fold equity and just scooping up the 3BB Villain has left out there. Therefore, Hero should take the hands that have enough fold equity + equity when called and shove these, folding the rest.

Imagine for a second that Hero held two useless cards that lost 100% of the time post-flop. This is of course impossible, but it helps illustrate. If Hero's cards were totally worthless, then his raw RFE would be his true RFE.

RFE = R / (R + PG)

RFE = 19.5 / (19.5 + 4.5)

RFE = 81%

Clearly shoving two blank cards then is a blunder. Hero's true fold equity shouldn't often be more than around 50% assuming that Villain opens about 20% from this seat and folds half of that to a shove.

This target can be brought down very easily indeed by Hero having some equity when called. What happens, given the assumption that Hero's fold equity is 50%, if Hero has **73o** and decides to put Villain all in? Will this be +EV? Let's find out how to calculate the EV of this shove. For simplification purposes, we'll make the clearly false assumption that the BB folds 100% of the time. In reality it won't be too far from this, but he will wake up with the nuts and interfere with our

objectives from time to time.

The EV of a Pre-Flop Shove

There are two branches to consider in a spot where Hero shoves over an open: the one where Villain folds and the one where he calls.

Branch 1: Villain folds

EV here is simply the amount by which Hero profits in picking up the pot. In this case, Villain's 3BB open + the dead blinds (including Hero's own blind).

EV When Branch 1 Occurs = 3 + 1.5 = +4.5BB

But Branch 1 does not happen all of the time, and so we need to multiply it by how often it occurs. In this case 50% of the time since Villain is assumed to be folding half of his range.

+4.5 x 0.5 = +2.25BB

Branch 1 EV = +2.25BB

Branch 2 : Villain Doesn't Fold

In this branch Hero loses his investment minus the amount he gets back from the pot at showdown based on his equity vs Villain's range. Against the top 10% of hands **73o** has a lowly but very influential 26% equity.

Hero invests 19.5BB and he gets back 26% of the total pot. The total pot will be two 20BB stacks + the dead BB and so Hero gets back 26% of 41BB.
Hero gets back **10.66BB**.

Hero's net loss when Branch 2 happens is 19.5 - 10.66BB = 8.84BB
Hero's EV when Branch 2 occurs is -8.84BB.

Again, we must multiply this by its frequency of occurrence.

-8.84 x 0.5 = -4.42BB

Branch 2 EV = -4.42BB

Now we simply add our two true EVs together to find the real EV of shoving the hand in question.

Overall EV = 2.25 + (-4.84) = -2.59BB

Shoving a terrible hand like **73o** here is -EV but not by a great deal!

Now we know the procedure, let's we plug in our actual hand of **55** and see what happens.

Branch 1 EV = +2.25BB. No change here.

It's Branch 2 that changes. **55** has 41% equity vs. the top 10% of hands. Hero now gets back 41% of the 41BB pot or 16.81BB. He only loses 19.50 - 16.81BB when called or 2.69BB.

His EV then is 2.69 x the 50% frequency with which the branch occurs.

Branch 2 EV = -1.35BB

Overall EV = 2.25 + (-1.35) = +0.9BB

With these assumptions in in place, it's profitable for Hero to put Villain all in, but does he want to do this?

No! The problem here is that unlike the last hand, this is not an end of action spot. In the real world, very occasionally, BB will wake up with a big hand and come over the top. In this case, Hero would be forced to lose 20BB if he folded. It's therefore more sensible here to just make a standard 3-bet size and accept that as far as CO is concerned it represents his stack. Even if CO makes the bizarre play of flatting Hero's 3-bet, the measly remainder of the money will go in on pretty much all but the very worst flops. Hero might even save a few bucks on **AsKsQs**.

Hero raises to 9BB

40BB - The 3-Betting Stack

Let's turn to an example where Villain has a more common stack size and one that short-stacking Regs like to wield aggressively. The reason that this stack can be played so aggressively pre-flop is that a smallish 3-bet can be used to create a very unpleasant situation for the opener. Take the following situation for example:

Hand 144

UTG – Unknown
HJ – Unknown
CO – Hero
BU – Reg (20/15/**11**) [40BB]
SB – Unknown
BB – Unknown

Dealt to Hero: X X

UTG folds, HJ folds, **Hero raises to 2.5BB**, **BU raises to 6BB**, SB folds, BB folds, **Hero?**

Hero is presented with two strategic problems facing the small 3-bet from this stack size.

- 4-betting any reasonable size commits Hero to the pot and so 4-bet/folding as a bluff is not an option.
- Calling offers very poor implied odds and so is less favourable than normal.

Hero's 4-bets now have to risk more money to take down the pot since they are effectively shoves, but shoving instead of 4-bet/folding as a semi-bluff comes with the bonus of realising equity, which a 4-bet bluff at 100BB stack depth usually does not have as Villain either folds, or shoves and we fold.

Hero's risk of 37.5BB to win 10BB here is pretty poor and leaves him with a raw RFE% (before adjustment for equity) of 79%. This is just about as unfavourable as shoving over the 20BB stack's open was in Hand 143. Nevertheless, Hero can profitably shove hands with reasonable equity. If we assume that BU 3-bets a wide 15% of hands in this spot and 3-bet/calls the top 7.5% of hands then again we have 50% fold equity. I've chosen 15% because his 11% is an overall average and BU vs. CO is a spot in which he should be exceeding that average.

Using the same hole cards of **55**, Hero now has 37% equity vs. Villain's 7.5% 3-bet/calling range of **[77+, ATs+ AJo+]**. Let's crunch the numbers and see if **55** is still a +EV shove.

When Branch 1 occurs, Villain folds and Hero's EV is **+10BB** (the total pot)

This branch happens 50% of the time so we multiply the above by 0.5.

Branch 1 EV = +5BB

In Branch 2, Hero has 37% equity vs. the top 7.5% of hands. Hero now gets back 37% of an 81.5BB pot or 30.16BB. He loses his investment of 37.5 - 30.16 = 7.34BB.

Branch 2 also happens 50% of the times so again we multiply by 0.5.

Branch 2 EV = -3.67BB

Overall EV of shoving = 5 + (-3.67) = **+1.33BB**

One mistake a lot of players make in this spot is to underestimate how good it is to shove. Missing +EV shoves here is one mistake I commonly try to eliminate in my students in these spots.

So Hero can defend quite wide by shoving, but how wide is too wide and what hands are better to call than to shove? Hero must make sure that he finds shoves with hands that are poor playability wise, but have reasonable equity when all in pre-flop. He should then seek to flat hands with lesser equity but higher good pair potential. Hero's required equity when called in order to shove profitably can be solved for algebraically as follows.

0 = Branch 1 EV + Branch 2 EV

0 = 10 + (81.5X - 37.5) Where X is Hero's equity when called.

0 = 81.5X - 27.5

81.5X = 27.5

X = 33.74

So It turns out that Hero needs **34%** equity vs. Villain's calling range to make a +EV shove, but that doesn't make shoving better than calling as calling may also be +EV; sadly how +EV can only be estimated.

Having decided which hands are best to flat and shove with, Hero should then fold hands that are insufficient in both of these departments. His overall defense strategy should look something like this:

AA	AKs	AQs	AJs	ATs	A9s	A8s	A7s	A6s	A5s	A4s	A3s	A2s
AKo	KK	KQs	KJs	KTs	K9s	K8s	K7s	K6s	K5s	K4s	K3s	K2s
AQo	KQo	QQ	QJs	QTs	Q9s	Q8s	Q7s	Q6s	Q5s	Q4s	Q3s	Q2s
AJo	KJo	QJo	JJ	JTs	J9s	J8s	J7s	J6s	J5s	J4s	J3s	J2s
ATo	KTo	QTo	JTo	TT	T9s	T8s	T7s	T6s	T5s	T4s	T3s	T2s
A9o	K9o	Q9o	J9o	T9o	99	98s	97s	96s	95s	94s	93s	92s
A8o	K8o	Q8o	J8o	T8o	98o	88	87s	86s	85s	84s	83s	82s
A7o	K7o	Q7o	J7o	T7o	97o	87o	77	76s	75s	74s	73s	72s
A6o	K6o	Q6o	J6o	T6o	96o	86o	76o	66	65s	64s	63s	62s
A5o	K5o	Q5o	J5o	T5o	95o	85o	75o	65o	55	54s	53s	52s
A4o	K4o	Q4o	J4o	T4o	94o	84o	74o	64o	54o	44	43s	42s
A3o	K3o	Q3o	J3o	T3o	93o	83o	73o	63o	53o	43o	33	32s
A2o	K2o	Q2o	J2o	T2o	92o	82o	72o	62o	52o	42o	32o	22

Shove Call Fold

Figure 80 - Dealing with 40BB 3-Bettors

This strategy has Hero folding 50% of his own range vs. the 3-bet. This does well to counter Villain's strategy as Villain's 3-bet folds have a raw RFE of 6 / (6 + 4) = 60%. We can deduct 10% from this target and probably no more since so many of Hero's defends come in the form of shoves, against which Villain realises no equity with his 3-bet/fold range. Hero is playing in such a way that he:

A: Is only shoving or calling hands that are +EV in a vacuum to do so.

AND

B: Negates any extra 3-bet folds that Villain adds to his range.

Hero is roughly balanced and protected vs. the 40BB stack as long as he finds all of these +EV

shoves. He must avoid overfolding the situation at all costs.

There are of course other exploitative adjustments Hero can make in order to make life harder on the 40BB stack. If he ups his open size from 2.5x to 3x, then BU will be offering Hero a much better price on a 4-bet shove and, perhaps paradoxically, Hero will be able to defend wider having opened bigger vs. this particular stack size. That said, he would lose more the time he folds and would be advised to accompany the larger sizing strategy with a tightening of his opening range.

If Villain's 3-betting was really out of hand, then Hero should make this tightening adjustment, open large and just shove a huge amount of his range to print money. The smaller Hero opens, the wider his calling range should be as his implied odds are likely to better depending on Villain's 3-bet size. He should also shove fewer hands if he opens smaller due to the worse risk reward ratio on a shove.

Overall, there are many ways that Hero can counter this strategy.

This section has been very instructive not just for countering the aggro 40BB stack, but also in teaching a general awareness of poker math in action. Make sure you're comfortable with the above theory before we move on.

Finally, it should be noted that if a 40BB stack opens, the situation is exactly the same with roles reversed. Hero can bear all of the above theory in mind when formulating his 3-bet range using the effective stack of 40BB; the last thing he wants to do is offer Villain the adjustment of shoving a very wide range. Have a go at building an optimal 3-bet game for this spot with the roles reversed. Try out various open sizes for Villain and see how Hero's strategy evolves.

60-80BB: The Awkward Stack

This stack size is one we'll commonly run into when facing weaker players. Fish will tend to buy in for random amounts, depending on what they feel like gambling that day, or on how much money is left in their account. They may also buy in for 100BB and then just not top up having lost a few pots. Hero should assume these stacks to be weaker players almost always. While there are Regs who buy in for these neither here nor there amounts, they're few and far between.

The first thing that should be recognised against these stacks is that it will be easier to get the money in post-flop. With a 60BB effective stack against a weaker player, value betting until all-in post-flop with decent top pair hands becomes much more attractive. As a result, Hero should favour opening, isolating, and 3-betting hands that can flop top pair frequently. At the same time, implied odds nut potential hands decrease significantly in value.

Let's take an example where the awkward stack size causes Hero to make a play that he wouldn't consider 100BB deep in many situations at all.

Hand 145

UTG – Unknown
HJ – Hero
CO – Unknown
BU – Unknown
SB – Aggro Fish (38/27/**14**) [64BB]
BB – Unknown

Dealt to Hero:

UTG folds, **Hero raises to 3BB**, CO folds, BU folds, **SB raises to 10BB**, BB folds, **Hero?**

With 100BB we'd be inclined to 4-bet/call for value vs. such a player. Our equity when all in would be very decent, and moreover, Villain could flat a lot of dominated hands out of position to the 4-bet. Hero would also have more room to utilise his positional and skill advantages to outplay Villain post-flop and would expect Villain to make a lot of mistakes. Shoving 100BB deep would not be the highest EV play unless Villain 3-bet and called off his stack very wide indeed. With these stacks, however, 4-betting to a smaller size completely commits Hero on all but the very worst of flops. Consequently, when Villain then flats the 4-bet he is making far less mistakes by stacking off post-flop and his position is less of a handicap.

AKo is not a hand that figures to be miles ahead of Villain's range, though it should be in good shape. 4-betting small allows Villain to realise equity for cheap, play fit-or-fold, and given how pot committed we are post-flop, a strategy of flatting the small 4-bet to continue with one pair or better should actually serve him quite well. We'd much rather simply shove at this stack depth for a few reasons:

- Villain loses the ability to mine for a pair on the flop and stack us when he hits one.
- Villain will likely felt much wider pre-flop with a 60BB stack than he would with 100BB so Hero's equity if Villain calls the shove is much better at the shallower stack depth, this makes shoving far less thin for value. Villain likely does not fold as many worse hands eg. **AJo**.
- Hero's shove protects his equity in the hand. Hero would much rather Villain folded a hand like **QJs** or **66** than took a flop with them given the significant and very large slabs of equity these hands have against **AKo** respectively.

Hero raises to 64BB (all in effective).

Now, how can we adjust our own 3-bet game to maximise our EV vs. an opener with 60-80BB?

Hand 146

UTG – Unknown
HJ – Unknown
CO – Unknown
BU – Tight Passive (24/13/70/64) [70BB]
SB – Unknown
BB – Hero

Dealt to Hero:

UTG folds, HJ folds, CO folds, **BU raises to 3.5BB**, SB folds, **Hero?**

BU is just the type of player Hero would love to 3-bet bluff a wide polar range against for fold equity purposes. Hero's value range will be narrow and his bluff range will span quite far down into his would-be folding range (hands too weak to call with.) This strategy is made even more lucrative by Villain's terrible cocktail of huge sizing and high **Fold to 3Bet After Raise** stat. This unique stack-size allows Hero to pick sizing that makes both shoving and 4-bet bluffing unattractive means of defence for Villain.

A standard 3x 3-bet to 10.5BB would be too large and would allow shoving to be a reasonable counter strategy. Hero wants to hurt Villain's RFE on a shove but still maintain a reasonable degree of pre-flop fold equity. For this reason he should err on the smaller side. A 3-bet of 8.5BB here should still mop up a reasonable amount of EV in fold equity and renders a shove by Villain a little ridiculous due to a poor risk:reward ratio. Note though that a smaller 4-bet to 20BB with the intention of folding to a shove is probably even worse since Villain will be pot committed and have to call off the rest anyway with most reasonable 4-bet bluffing hands.

Hero will make a great deal of money in this spot and needs to attack it aggressively with smallish sizing.

Hero 3-bets to 8.5BB.

14.3 Playing Deep Post-Flop

In this section we'll be focusing less on bands of stack-size and more on the themes of being deep post-flop. We'll start out with one of the most misplayed areas of deep stack poker.

Out of Position Deep - Keep Pots Smaller.

Conventional poker wisdom states that calling is generally more profitable in position than out of position and so, whether Hero is looking for fold equity or seeking to extract value, he is advised to have more of a raising game post-flop on the flop and turn where he is out of position. One exception to this rule is when stacks are deep. As we know already, deeper stack-sizes exaggerate positional advantage. Another thing they do is increase reverse implied odds. When Hero builds a big pot 100BB deep, he can typically expect to be in excellent shape when a lot of money goes in with the second or third nuts (depending on board texture). Should 250BB end up in the middle, however, without an aggressive dynamic, the average Reg will have a much tighter range and bottom set may suddenly feel like a bluff catcher that has just value owned itself for two and a half stacks!

Being out of position removes Hero's ability to control the size of the pot on later streets. He cannot prevent Villain from betting the pot having checked and may find himself in some very uncomfortable situations should he build very large pots with unnutted hands. As we are always concerned with our range against competent Regs and not just our two cards in a vacuum, we'll often decide to forego having a raising range altogether when deep on the flop out of position, as there are so few (if any) combos we are comfortable building an enormous pot with. Balance, as always is very important where applicable.

Have a look at the following situation:

Hand 147

UTG – Unknown
HJ – Reg (20/16/55/50) [270BB]
CO – Unknown
BU – Unknown
SB – Unknown
BB – Hero

Dealt to Hero:

UTG folds, **HJ raises to 3BB**, CO folds, BU folds, SB folds, **Hero calls 2BB**.

Flop (6.5BB, 2 players)

Hero checks, **BU bets 5BB**, **Hero?**

100BB deep, this spot is a raise. The flop is wet enough that we want to get our value fast and we have range advantage due to having a higher concentration of sets and straights than Villain has. While both players can have these hands, they are less diluted in our range than they are in Villain's. If we get 3-bet on the flop with a 100BB stack, we'll simply shrug and shove expecting to have enough equity vs. a stacking off range of two pair plus and good draws (maybe even with the odd overpair that wants to protect itself vs. draws).

Getting 3-bet on this flop 270BB deep is extremely unpleasant. We know that this is often far from the last bet we'll be facing and we've created a situation where Villain can pile on pressure on any run out, even ones that seem safe for us. Bloating this pot with such a deep effective stack out of position is horrible. Even if Villain just calls our poor flop raise, he will have the option of injecting the pot further on the turn and river should we need to start slowing down. If we raise and the turn is the **Qd** we're forced to check and decide between folding or calling and drawing to a hand that could be

crushed by bigger sets. The reverse implied odds are awful.

Even on a 'safe' turns like the **4h** we may feel uneasy about betting and getting raised. Again, if we get raised on the turn with 100BB stacks we'll be gladly calling off the rest.

But if we just call flop aren't we letting Villain see free cards cheaply on a wet board? Yes, but don't be so quick to assume that that's the end of the world. Villain's range is far from just draws, if it were we'd simply shovel money into the pot then check/fold if they get there, check/calling all blank rivers. Sadly, his range contains better nutted hands too and even the draws he does have are very happy calling a flop raise in position with great implied odds due to stack-size.

Building big pots out of position with unnutted and/or very vulnerable hands is a recipe for getting destroyed. We don't need a raising range here as our positional disadvantage demands that we keep the pot smaller so as to cut down on our reverse implied odds. There's only so much we can lose to better hands by check/calling down and we can often still get three streets of value from an overpair on good run outs this way. I don't think it's necessary to deploy any kind of raising range here at all, but if you do want one, make sure that it's constructed with totally nutted hands, nutted draws and hands that are closer to pure bluffs. This will make life much easier on future streets out of position.

Hero calls 5BB

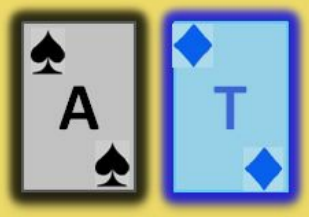
In Position Deep - Build Away

In position, the worries of the last hand are relieved. We now have the magical weapon of pot control. Say we raise up the flop in the last hand, get called and the turn comes a nightmare card. Well, most Villains will continue to check to the aggressor; after all our range is far from capped and we could have the nut flush on a diamond turn very easily. In fact, Villain should generally range check every turn to the aggressor. Now we can check behind and go to the river. We may have a difficult decision to make here should Villain bet out, but at least it's just one difficult decision in a controlled pot. We're not check/calling our huge stack away with no idea as to Villain's bluffing frequencies.

Also, in position we get to play the role of the deep-stacked bully. As our positional advantage is so huge, we can apply pressure from an early street and always bail into a free card later; if that is, we don't want to blast away and make Villain's head hurt with two pair. Look at the following hand:

Hand 148

UTG – Unknown
HJ – Unknown
CO – Reg (20/16/55/50) [270BB]
BU – Hero
SB – Fish (56/5/0/25)
BB – Unknown

Dealt to Hero: 

UTG folds, HJ folds, **CO raises to 3BB**, **Hero calls 3BB**, SB folds, BB folds.

Flop (7.5BB, 2 players)

CO bets 6BB, **Hero?**

Hero's pre-flop call would be somewhat questionable if the Fish had not been in the SB. As is, Hero really wants to get into pots in position to weak players with a dominating hand so this is totally fine. SB folds: no cigar.

So now we find ourselves heads-up on the flop with a gut-shot, overcard and a beautiful blocker to the nut flush. Why is this latter feature so important deep? It stops Villain from being able to turn the nuts and this allows us to apply a lot of pressure on a spade turn. The extra equity from the gutter makes life even happier and this is clearly an excellent part of our range to turn into a bluff on a flop where we can have all the best value combos going. I'm not at all opposed to having a raising range on this flop. It should still be very polarised, however, and I'd probably stick to something like **[88 JJ T9]** for value, some nut flush draws that thrive with this stack depth and some weaker bluffs to balance. We can always pot control by checking back the sets and straights on spade turns and can apply lots of pressure with our nut flushes and bluffs on those turns. Note that we're not raising this flop with a **Q**-high flush draw - it prefers a less than enormous pot.

Building the pot sooner in position when deep allows us to wield a giant hammer over our opponent and create a lot of discomfort. As a result, and due to this flop smashing our range really well, CO should deploy a strategy of checking most of his range on this texture. Few players actually do this though until you get to the higher and tougher limits. The brain dead Regs who auto c-bet like monkeys here deserve the beating they're likely to receive.

Hero raises to 24BB.

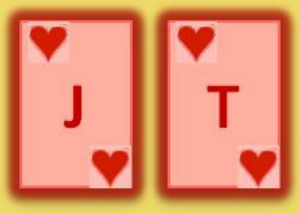
Woah, huge sizing! Absolutely! Hero has all three streets to apply pressure and is planning on barreling lots on many run outs. Therefore, he is perfectly happy building a huge pot with all parts of his range. The value wants to win the maximum and is close to nutted and the bluffs don't need a good RFE right now as they'll be blasting away so often on turns and rivers with increasing fold equity. This sizing makes a lot of sense - maximise the EV of your positional deep stack hammer!

Deep Stacks and Draws

The deeper we get, the greater implied odds become, but the more nutted our draws need to be to realise them without huge downside. There are occasions with 100BB effective stacks on which folding a weak draw on the flop or turn is totally fine, but where we must call with 200BB stacks.

Hand 149

UTG – Unknown
HJ – Unknown
CO – Whale (64/34/0/25) [200BB]
BU – Reg (27/21/56/54)
SB – Hero
BB – Unknown

Dealt to Hero: 

UTG folds, HJ folds, **CO raises to 2BB**, **BU raises to 7BB**, **Hero calls 6.5BB**, BB folds, **CO calls 5BB**.

Flop (22BB, 3 players)

Hero checks, **CO bets 8BB**, **BU calls 8BB**, **Hero calls 8BB**.

Turn (46BB, 3 players)

Hero checks, **CO bets 10BB**, BU folds, **Hero?**

Pre-flop is a clear cold call. It's not that we want to play this hand 100BB deep OOP against the Reg's likely linear 3-bet value range that dominates us, it's that we want to get into a pot 200BB with a whale who is extremely likely to be calling the 3-bet. Our hand thrives from implied odds rich multiway situations with these stacks so calling should be obvious.

Now onto post-flop, but first remind yourself of the difference between floating and chasing, if there is any doubt.

This whale offers very little in the way of fold equity and so the plan for Hero is simple: chase and draw his way to victory in a passive fit-or-fold manner. This is winning play against this kind of opponent. So Hero calls the flop because he thinks that the 8.5% of the time he'll turn the nuts is enough to justify the investment. He'll need to win back almost twelve times this investment from the pot and his opponent's stacks combined for the call to be profitable. To find this just turn our 8.5% chance to improve on the next card into a fraction by dividing both sides of the fraction by 8.5.

8.5/100 = 1/12.

Hero's investment is 8BB.

12 x 8 = 96 and so Hero needs to claw back 96BB to break even on the flop call.

After calling the flop, the pot will then stand at 46BB. This leaves another 96 - 46 = 50BB to be won over the turn and river. Against this opponent Hero decides this will be an easy task and there's always the chance of making more from the Reg too and so the flop is a very easy call. There's also the prospect of a free river card from time to time as well when Hero misses the turn and this helps implied odds further by allowing the realisation of more equity.

On the turn Hero must call 10 into 46. His chance to improve on the next card is now 4/46 = 8.7%.

8.7/100 = 1/11.5

Hero needs to win 11.5 times his investment on average to call the turn. He needs to make 10 x 11.5 = 115BB.

There will already be 66BB in there should Hero call the turn. This leaves a residual average target of 115 - 66 = 49BB more to be won on the river should Hero get there. Again this seems very reasonable against our whale opponent and so Hero should continue to chase.

If we make stacks shorter, the average payout will be significantly less and all streets will suffer a hit in implied odds. Chase more without fold equity as stacks get bigger, particularly where Villain has a preference for the call and raise buttons.

Hero calls 10BB

Deep Stacked Variance

Winning a 500BB pot can be exhilarating. Losing one can be crushing. Playing deep generates a lot more variance than we are used to dealing with in 100BB play. For this reason, I recommend that Hero takes extra care in avoiding -EV deep situations unless other table conditions are extremely favourable. With an aggressive and competent player to act after him, and where the effective stack is deep, Hero should acknowledge that his positional and possible skill disadvantages are inflated. Pride is meaningless in poker. Unless there are multiple Fish or a whale at the table, I advise Hero to get up and find a new seat. The Reg to his left has full reign to make life very unpleasant and create

treacherous and -EV situations by the dozen if he knows what he's doing.

Similarly, when Hero is in position and deep, especially vs. inferior players, he should do everything possible to remain at the table and capitalise on the extra EV for as long as possible. Whether it means holding off on dinner and eating crisps for the next two hours, or losing an hours sleep, it might well be worth it, though the reader should of course maintain a healthy poker/life balance and avoid playing when exhausted or overly emotional.

For the beginning player, even one who's reached this part of The Grinder's Manual, it may be necessary to avoid deep stack play except in very clearly favourable situations at first. It's true that if you don't practice you won't get better, but save practice for obviously +EV spots. Don't stay there 300BB deep vs. an aggressive player if you can already feel it affecting your game and emotional composure. As you improve, you'll become better and better at detaching yourself from stack size emotionally, but attaching yourself to it logically and changing your game in the appropriate way.

Finally, if you're going to play in games where you're frequently deep stacked like Zoom without a ***get up and come back*** strategy for getting rid of your deep stack, you'll need a larger bankroll. I won't discussing bankroll management here as this is a technical manual about poker strategy, but there's a lot of great information floating around on the internet. Go find it.

14.4 Playing Shallow Post-Flop

Room To Check

In Chapter 5 we talked about times when betting out and pot building from the get go was not the best strategy for extracting value, even as the pre-flop raiser. Many of these situations occurred when the effective stack size was shallow post-flop. Having the room to check a flop or turn means that there is no necessity to bet that street in order to ensure stacks can go in at some point should Villain call. With this in mind, take a look at the following situation in a 4-bet pot and think about how Hero wants to play the different parts of his range. His range for 4-betting pre-flop is stated in the hand history.

Hand 150

UTG – Unknown
HJ – Unknown
CO – Unknown
BU – Hero
SB – Unknown
BB – Reg (19/15/**10**)

Dealt to Hero: [JJ+, AK, K6s-K8s, A2o-A5o]

UTG folds, HJ folds, CO folds, **Hero raises to 2BB**, SB folds, **BB raises to 7.5BB**, **Hero raises to 18BB**, **BB calls 10.5BB**.

Flop (36.5BB, 2 players)

BB checks, Hero?

Hero is using a fairly bluff heavy polarised and blocker filled 4-bet range, presumably because he thinks fold equity is forthcoming. Villain is uncompromising enough to flat the 4-bet and so to the flop we go.

There are two important things to note about the resulting situation on the flop:

- Hero needs just two streets of reasonable bet sizes or three of very small bet sizes to get the money in.
- Most of Hero's range does not want to get the stacks in and is some kind of moderate to weak SDV.

This points us towards one of two balanced strategies being best.

The first option is to check everything on the flop. This is in no way exploitable as Hero has many combos he can call the turn with like **[AA AK A2o-A5o]** and some pocket pairs too. He can call the **Ax** portion of these (or raise them) on the river should Villain bet again and fold the **[JJ-KK]** and **[A2o-A5o]** combos. If Villain checks again on the turn as he often will in practice (Hero should still be perceived to have range advantage in the eyes of a competent player) then Hero can check again with his pocket pairs, bluff his **[K6s-K8s]** and value bet his **[AA AK]**. He can split his wheel aces between bets and checks to keep his range fairly balanced and give him some more value combos. Then on the river, if called, he should shove his **[AK AA]** (15 combos) for a pot-sized bet. Villain will then need 33% to bluff catch and so Hero should bluff the amount of his **[K6s-K8s]** that gives Villain this much equity. 8 combos will do nicely and so Hero can bluff **[K6-K7s]**. There's really not much to choose from here, but **K8s** blocks **88**, which is not an impossible hand for Villain to get to the river with. We don't want to block **88** when we're bluffing; if it somehow got here, it's almost certainly folding.

The second option is to c-bet the entire range small enough on the flop to leave two more small to reasonable sized bets for the turn and river. Hero could then split his range similarly to the first strategy by polarising his turn betting range and checking back his SDV and calling the **[A2o-A5o]** chunk to a river bet. He would bet his air as a bluff and his value hands, well, for value!

The main point here is that regardless of Hero's chosen strategy, he has no need to bet any significant size on the flop with any of his range. Value and fold equity can both be gained to a greater extent later in the hand and the texture makes Hero's value hands extremely invulnerable. There is simply no urgency at all here. If we were deep, however, then checking the flop behind could be a big mistake as we'd often miss out on three streets of value as every street would demand a decent-sized bet in order to extract maximum value.

Hero either checks or bets 7BB

Stacking Off Lighter

It may sound obvious, but when there's less money left behind, Hero should be inclined to lessen his requirements for putting that money in. Let's take a closer look at a very common type of situation, which illustrates this necessity.

Hand 151

UTG – Hero
HJ – Unknown
CO – Unknown
BU – Passive Fish (34/8/**2**) [30BB]
SB – Unknown
BB – Unknown

Dealt to Hero:

Hero raises to 3BB, HJ folds, CO folds, **BU calls 3BB**, SB folds, BB folds.

Flop (7.5BB, 2 players)

Hero bets 3BB, **BU raises to 27BB (all in)**, **Hero?**

If this passive player was to make such a large raise size, or even a smaller one with standard 100BB stacks, Hero would be rather terrified to say the least. The all-in shove however usually sends an entirely more welcoming message. While the raise with deeper stacks from this player type typically means something to the effect of:

"I have the nuts and am terrified you might suck out on me

The all-in with the shallower stack often says:

"I have a pair and or a draw, or possibly a big hand, but I may as well just shove because it's all going in anyway."

The shove is far more depolarised and is not weighted towards hands that crush us. The large raise with deeper stacks is a sign that **88** is crushed way too often and is very rarely better than a coin flip.

Hero should fold in that case but call this shove happily.

Hero calls 24BB

Note that Hero's flop bet is too small. He is probably trying to induce some action and get the money in over three streets, but vs. a passive player who will be more inclined to call than raise, he should just pot the flop and shove the turn and get that value while it's there - this is a board that gets ugly fast and kills value on later streets.

Playing Draws Shallow

We know full well by now that when there's less money left behind, implied odds are slashed. This means that we are much better off in many shallow stacked cases treating our draws as semi-bluffing weapons. We often have the luxury of being able to shove all-in, capitalising on any fold equity while at the same time ensuring we'll get to see all five cards and realise our full equity in the hand.

The Villain in Hand 152 below is not someone we'd typically consider bluffing against on the later streets, but Hero's turn play must be aggressive.

Hand 152

UTG – Unknown
HJ – Unknown
CO – Unknown
BU – Hero
SB – Aggro Fish (50/35/**7**) [18BB]
BB – Unknown

Dealt to Hero:

UTG folds, HJ folds, CO folds, **Hero raises to 2BB**, **SB calls 1.5BB**, BB folds.

Flop (5BB, 2 players)

SB checks, **Hero bets 4BB**, **SB calls 4BB**.

Turn (13BB, 2 players)

SB checks, **Hero?** (12BB effective)

There is some chance of fold equity on this card even against a short stacked Aggro Fish and Hero needs very little for shoving to be correct given that:

- Hero has a lot of pot equity. Against a typical non-folding hand like **QdTd** he has 34%. This means that if Hero were to check, he would be forced to check/call since he meets RE, and of course shoving and picking up fold equity to boot is an improvement over that line.
- Villain can even call off worse draws, which Hero utterly dominates. Fish frequently make this mistake when their stack has reached "who cares I'm too short to fold" territory.

Note that, with deeper stacks Hero would have to fear being raised and may want to check/call the turn depending how little fold equity he expected to have vs. the Fish in question.

Hero bets 12BB.

Wrapping Up

Well, that brings us to the end of the technical topics. That doesn't mean you're now an expert and I advise the reader at this point to be far from done with The Grinder's Manual. The examples can be revised multiple times and it will take many revisits to fully incorporate the material in this manual into your game. You have the tools right here, but you will not be able to wield them all competently without a lot of practice. Be patient, focus on improvement and the monetary results will come in time. It's not your job to force that part so be patient and keep working hard on your game.